before measurement $\dfrac{(\alpha+\beta)|00\rangle + (\alpha-\beta)|11\rangle}{\sqrt{2}}$

`lift`, then V

record 1 **(safe,** $g = |0\rangle$**)**

$$\frac{\alpha+\beta}{\sqrt{2}} \cdot |0\rangle \otimes |0\rangle$$

weight $|\alpha+\beta|^2/2$

record 0 **(unsafe)**

$$\frac{\alpha-\beta}{\sqrt{2}} \cdot \frac{|01\rangle + |10\rangle}{\sqrt{2}}$$

weight $|\alpha-\beta|^2/2$